PRODUCT DOCUMENTATION · MOPR11 · VOICEBRIDGE CTI FOR SALESFORCE

What VoiceBridge actually is, what is finished, and what is not

A pictorial walk-through of the product as it stands at code version 0.28.0. Five diagrams, each stating the status of every part in words as well as colour, followed by the completed and not-completed registers and the reason behind every gap.

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Author / prepared by	Claude for Senthil (KP)
Owner / approver	Senthil (KP) — product owner and sole decision-maker on every open item
Describes	voicebridge-cti **v0.28.0**, package hash `f0f4eb4ab67d`, released 20 Sep 2026 10:45
Classification	Internal — contains an honest completion assessment not intended for a prospect or a listing
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Review inputs	External review V001 (ChatGPT, 13 Sep, REJECT) · Perplexity review V001 (16 Sep, HOLD) · five internal cold-review passes

1 · Document control

How to use this section. One row per issue of this document. Amend the row, never the history: a new fact about the product means a new version of this file, released alongside a code zip and a tracker version so the three always describe the same build.

VERSION	DATE	AUTHOR	WHAT CHANGED, AND WHY	STATUS
v001	20 Sep 2026 12:32	Claude for Senthil/KP	First issue. Created because the written trackers describe the build accurately but do not let a cold reader *see* where the product stops. Describes v0.28.0 exactly as shipped and verified from its own zip; every status claim traces to a test, a gate, an operator step or a named decision.	Released
v002	—	—	*Planned.* Opens when the S13 decision gate closes (D-V8 strategy ruling, D-V9 listing confirmation), or when the first operator step produces live evidence — either changes the picture in figures 1, 3 and 4 materially.	Planned

Naming convention. `claude_v<NNN>_<product ref>_<document name>_YYYYMMDD_HHMI.<ext>`, with the timestamp supplied by Senthil at release. The same figures are also delivered as standalone SVG and PNG files for reuse in decks and in the workbooks.

2 · Context — what this product is, and how it got here

How to use this section. Read it once before the diagrams. It carries the six facts that make the pictures legible; everything after it assumes them.

The product. VoiceBridge CTI is a Salesforce AppExchange application for mid-market B2B sales teams: a softphone that lives in the Salesforce utility bar, click-to-dial from any record, automatic logging of every call (and, in the pitch, every WhatsApp thread) back onto the Contact or Lead, plus lightweight cadences, coaching scorecards and Flow actions. It is product P02 of a twelve-product marketplace portfolio, built solo.

How it was built. Fifteen numbered code releases between 13 and 20 September 2026, each one packaged, re-extracted and re-verified from its own zip. Every release is gated on 100% statement, branch, function and line coverage with no coverage pragmas — a standing portfolio rule (D13) — and on a release-evidence pack bound to the package hash.

What three reviews found. Five internal cold-review passes found roughly forty defects inside code that already had 100% coverage. An external review (ChatGPT, 13 September) returned **REJECT** with fourteen findings, five of them blockers. A second independent review (Perplexity, 16 September) returned **HOLD — re-platform or pivot**, with six further findings and one strategic fact nobody had recorded: the Salesforce API the whole in-org surface depends on is being retired.

Where that leaves the build. Version 0.28.0 closed five of the six Perplexity findings and four worse defects that the fixes themselves exposed. The code is now, by any reasonable reading, an engineering-complete service — and it has still never placed a telephone call, never written a record into a real Salesforce org, and never run against a real database. Those two sentences are both true at once, and holding them together is the point of this document.

The vocabulary used on every diagram

STATUS	WHAT IT MEANS HERE — THE TEST APPLIED TO EVERY SINGLE BOX
Built and proven	The code exists and the evidence exists *today*, in a form you can re-run yourself: an automated test, a build gate, a database check, or the release evidence pack. Nothing in this state depends on a system we do not have.
Built, unproven live	The code exists and is fully tested, but every counterparty in the test is a double we wrote. It has never met a real Salesforce org, a real PostgreSQL server, a real carrier or a real browser inside a real utility bar. This is not a quality judgement on the code; it is a statement about what has been observed.
Not built	The component does not exist. In every case the reason is a decision that has not been made or a commercial relationship that does not exist — never an unsolved technical problem.
Not ours	Salesforce's, a vendor's, or the operator's. Shown because the mechanism is unreadable without it.

Each state is written on the box as well as coloured, so the diagrams survive greyscale printing and do not depend on colour vision.

3 · The product map

Use this figure when someone asks "what *is* this thing?", or when you need to show that a change in the left-hand column (the Salesforce surface) does not threaten the middle and right ones. Read it left to right: the customer's org, our service, the

outside world.

VoiceBridge CTI — what the product actually is, and where it stops

Every box states its own status in words. Dashed red = the component does not exist; the reason is on the box. Code v0.28.0, 20 Sep 2026.

Built and proven now (tests, gates, evidence pack) · Built and tested, never run against a live org/database/carrier · Not built — blocker named on the box · Not ours (Salesforce, a vendor, or the operator)

1 · SALESFORCE ORG — THE CUSTOMER'S

Lightning utility bar
Where the rep works all day. Hosts our panel in a frame.
SALESFORCE'S — we only supply the URL it loads

Softphone panel — widget/softphone.html
Dial pad, live call state, outcome + notes.
Trades the install cookie for a bearer, then calls us.
FIXED v0.28.0 (F-P01): the page had never been served at all

Record page — click-to-dial
Any phone field becomes a dial link via onClickToDial.
BUILT · needs a live org to see it work (O7b)

CallCenter definition (Open CTI)
The metadata that points the utility bar at our panel.
Ships in the 2GP package with the permission sets.
BUILT · the API underneath it retires 28 Feb 2028 — see below

Permission sets × 3 (owner / admin / member)
Generated from the same role matrix the server enforces.
BUILT · install shape untested in a real org (LC-5)

Task + VoiceBridge_External_Id__c
One activity per call. Our external id is the upsert key,
so a retry updates rather than duplicating.
BUILT · no Task has ever been written to a real org (O7)

Contact / Lead
Matched from the phone number so the call attaches
to the right person — or to nobody, honestly.
THE CUSTOMER'S DATA — we only read and attach

External Client App (OAuth)
Install + token refresh. Connected Apps are gone.
BUILT · never completed against a live org

1 install cookie → bearer

2 click-to-dial → POST /dial

4 REST upsert — one Task per call, keyed on our external id; answered by a test double, never by a real org (O7)

2 · VOICEBRIDGE SERVICE — OURS (NODE / EXPRESS 5)

INBOUND — A REQUEST ARRIVES

Edge — every request
CSP + frame-ancestors, HSTS, 100 kB cap,
per-org and per-ip rate limits, request ids.
NEW in v0.28.0 (F-P04)

Identity boundary
Signed session with a jti and an epoch;
logout, revoke-all, role change kills sessions.
PROVEN BY TEST — 613 tests, 100% coverage

HTTP routes
/widget /dial /log /events /crm /ui /auth
/roles /blocklist /audit /metrics /status
PROVEN BY TEST — contract tests drive the real app

Feature modules
dialer · call logger · cadence · coaching
flow actions · billing · analytics · support
PROVEN BY TEST — business rules, not integrations

Number guard — before any dial
E.164 + 15 country rules, org blocklist,
do-not-call check. A refused number never dials.
NEW in v0.28.0 (VB-E-025)

Audit trail
sha256 hash chain, append-only in the database,
verifiable by the customer at /audit/verify.
NEW in v0.28.0 (VB-E-081)

OUTBOUND — WE CALL SOMEONE

Telephony seam
The contract an adapter must satisfy, plus a
simulator that labels every call simulated.
SEAM BUILT — the adapter behind it does not exist

CRM write-back — sfRest
SOSL phone match: one, none, or ambiguous.
Ambiguous → written unattached, never guessed.
BUILT · answered by a double, not Salesforce (O7)

Outbox + drain worker
A failed write queues and is retried by a
leader-elected worker; dead-letter after 8.
NEW in v0.28.0 (F-P03) · live proof at O10

Observability
Structured logs, RED metrics per route,
/metrics for a scraper, error reporting.
NEW in v0.28.0 · nothing scrapes it yet

SLO + burn alerts
The published 99.5% as code, with rules
an alerting backend can import.
BUILT · never fired in anger (game day O10)

Deploy + DR tooling
Blue/green with migration and health gates,
automatic rollback; pg_dump restore drill.
BUILT · no platform to run it on yet

PostgreSQL — 10 ordered migrations
Row-level security per tenant · tokens encrypted at rest (AES-256-GCM) · hash-chained audit
outbox + dead letter · worker leases · rate-limit windows · sequences · PKCE state
Every org-scoped statement runs inside a transaction that sets the tenant context.
BUILT · HAS NEVER RUN AGAINST A LIVE POSTGRES (operator step O6)

3 place the call

→ /events/ingest

3 · OUTSIDE — CARRIERS, PLATFORMS, VENDORS

Telephony carrier
The thing that actually rings a phone.
Twilio / Plivo / Exotel / Service Cloud Voice.
BLOCKED: D-V5 — no provider chosen, so no calls are placed

WhatsApp Cloud API
Send and receive business messages.
The 'omnichannel' half of the pitch.
BLOCKED: D-V6 — the claim is withdrawn until it is built

KMS / secret store
Unwraps the token encryption key at boot.
DEPLOY-TIME WIRING — the key ring in code is ready

Metrics + alerting backend
Scrapes /metrics and runs the burn-rate rules.
NO PLATFORM YET — rules are exported and waiting

Hosting platform + managed Postgres
Where the middle column would actually run.
NOT DEPLOYED ANYWHERE — this is why so much says 'unproven'

AppExchange listing
Security review, licensing, the public listing.
$999 per submission, 5–9 months lead time.
BLOCKED: D-V8 strategy, D-V3 billing route, ISV agreement

The one fact that governs the whole left-hand column — R-V21

Open CTI — the API the utility bar, the CallCenter definition and our softphone adapter all depend on — is in maintenance mode.
Salesforce ends its life on 28 February 2028 and has closed it to newly created Agentforce Service orgs (verified 20 Sep 2026, help article 005306437).
Columns 2 and 3 survive a move to Salesforce Voice with Partner Telephony untouched. Choosing that move is decision D-V8; confirming the listing rule is D-V9.

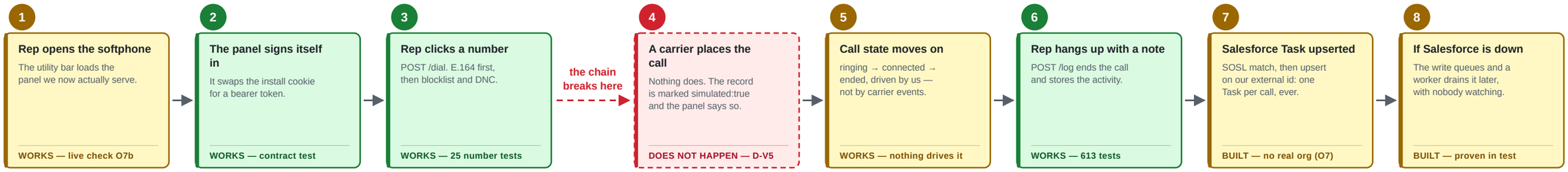
Figure 1 — The product map. The customer's Salesforce org on the left, our service in the middle, the outside world on the right. Three boundary crossings carry the whole story: the org reaches us and that works (arrows 1 and 2); we reach a carrier and there is nothing there (arrow 3, dashed); we write back to Salesforce and that is built but has only ever been answered by a test double (arrow 4). The red strip beneath is the fact that governs the entire left column.

4 · One call, end to end

Use this figure when the question is "does it work?" — because the honest answer is "seven of the eight steps do". It is also the fastest way to explain why the product is not simply "nearly finished".

One call, start to finish — and the single link that is missing

The chain is intact everywhere except step 4. Nothing downstream of it has ever been driven by a real telephone network.



What a customer would actually get today, if everything else were deployed

A softphone panel that opens, authenticates, validates and refuses the right numbers, records the call, writes one Task per call to Salesforce and survives a Salesforce outage — with every call marked "simulated: no telephony provider is connected, so +91… was not dialled" on the panel itself.

Why step 4 is empty, and what it costs

Choosing a carrier (D-V5) sets per-minute cost, and therefore the pricing floor in billing.ts — so it is a commercial decision, not a technical one. Building the adapter afterwards is one file against an existing contract, plus one inbound webhook route whose signature scheme is provider-specific.

Figure 2 — One call, end to end. Eight steps. Seven of them run today. Step 4 — a carrier actually placing the call — is empty, and because it is empty every call downstream is recorded honestly as simulated. The two notes below say what a customer would really get today, and why the empty step is a commercial decision rather than an engineering one.

5 · Completion, capability by capability

Use this figure in a status review or with an investor. The two chip columns are the whole argument: the code column is mostly green, the evidence column is entirely red, and one operator week would change that more than another month of building.

Completion by capability — what is done, what is only written down, and why

Two different questions: is the code there, and has anyone ever seen it work outside a test? For this product they have very different answers.

7	capability areas complete in code, with tests and gates behind them
2	partial — the code is there, the commercial inputs are not
3	not built at all — each blocked on one named decision
0	areas with any live evidence — nothing is deployed and no dev org exists

CAPABILITY — WHAT A BUYER WOULD ASK FOR	CODE	LIVE EVIDENCE	WHAT IS MISSING, AND WHY
Install, identity and permissions	Complete	None	OAuth on an External Client App, PKCE, signed sessions with logout and revoke-all. No install has ever completed against a live org (no dev org exists yet).
In-org softphone surface	Complete	None	The page is finally served (F-P01 — it never was). Whether the panel renders inside a real utility bar is operator step O7b.
Placing a real call	None	None	No telephony adapter. Blocked on D-V5: the provider choice sets per-minute cost and the pricing floor, so it is Senthil's commercial call.
Inbound calls and screen pop	None	None	Needs the same adapter plus a provider-specific webhook (signature scheme and event vocabulary differ per carrier). Blocked on D-V5.
Auto-logging to Salesforce	Complete	None	SOSL match and Task upsert on an external id, with a durable outbox behind it. Answered by a recorded double; live proof is O7.
WhatsApp / omnichannel	None	None	Never built. The claim was withdrawn from the help centre in v0.28.0 rather than left published. Blocked on D-V6.
Sales features (cadence, coaching, flows)	Complete	None	Business rules are complete and tested. Coaching counts events rather than reading transcripts — that needs the AI layer, which needs recordings.
Billing, plans and pricing	Partial	None	Trial, proration, dunning, limits, invoices all built. The prices themselves derive from D-V5, and the route to charge (Checkout vs ISVforce) is D-V3.
Durable storage and tenant isolation	Complete	None	Ten migrations, row-level security enforced by Postgres, encrypted tokens. Never run against a live Postgres — operator step O6.
Security hardening and audit integrity	Complete	None	Headers, rate limits, encryption at rest, hash-chained append-only audit. The external half is a penetration test, which needs a deployment.
Operations — observability, SLO, deploy, DR	Complete	None	Logs, metrics, burn-rate rules, blue/green with rollback, a restore drill. Nothing is deployed, so nothing has ever been operated.
Go to market — package, listing, pilots	Partial	None	Package source, listing copy and assets are drafted. Blocked on D-V8 (which platform), D-V3 (billing route), the ISV agreement and 5-9 months of lead time.

Read the two columns together

Nine of the twelve rows are complete in code and empty of live evidence. That is not an accident of sequencing: there is no dev org, no deployed

service and no database, so no amount of further building moves those cells. Operator steps O6, O7, O7b, O9 and O10 are what change them.

Figure 3 — Completion by capability. Twelve capability areas, each with two separate answers: is the code there, and has anyone seen it work outside a test. Seven areas are complete in code, two partial, three not built — and not one has any live evidence, because nothing is deployed and there is no developer org. The right-hand column names the specific blocker for each row.

6 · Release history and what comes next

Use this figure to show velocity honestly. Above the line is what was shipped and verified; below it is what cannot start, with the specific blocker on each card.

Where the code has been, and where it can go next

Fifteen numbered releases between 13 and 20 September alone, then a hard stop. Everything below the line waits on a decision, or on something that has to exist in the world first.

SHIPPED — EACH ONE VERIFIED FROM ITS OWN ZIP

v0.14 – v0.16
Surface and identity
The 7 missing screens, the OAuth controls Salesforce made mandatory, and the identity boundary that stopped self-named callers.

v0.17 – v0.22
Persistence, in stages
Twenty modules moved onto an async repository one green build at a time, then the legacy in-process store was deleted outright.

v0.23 – v0.24
Concurrency
Serial guards for check-then-write, then the dedupe pushed down into a Postgres unique index so two instances cannot both win.

v0.25
First external review
ChatGPT's review: REJECT. Nine of fourteen findings closed, including three of five blockers, in one release.

v0.26 – v0.27
CRM write-back, honest dial
The Task upsert that had never been written, and the end of the pretence: every simulated call is now labelled on every surface.

v0.28 ← you are here
Second external review
Perplexity's six findings closed, plus four worse ones the fix exposed. Hardening, encryption, RLS, audit chain, outbox worker.

20 Sep 2026

NOT SHIPPED — THE BLOCKER IS NAMED ON EVERY CARD

v0.29
Decision gate
No feature code. The six S13 decisions, the model rebuild, reclassification.
blocked: D-V4 … D-V9

v0.30
Dial path 1
The provider adapter and the inbound webhook.
blocked: D-V5

v0.31
Dial path 2
Inbound routing, screen pop, call-quality telemetry.
blocked: D-V5

v0.32-33
Partner Telephony
Connector, Contact Center, Voice Call records, 2GP.
blocked: D-V8 + dev org

v0.34
Live durability
Conformance against a real Postgres; cross-hop traces.
blocked: O6 + a platform

v0.35
WhatsApp
Cloud API, opt-in ledger, 24h window, templates.
blocked: D-V6 + Meta

v0.36
AI layer
Transcription, summaries, real coaching scorecards.
blocked: recordings first

v0.37
Enterprise close-out
SSO decision, pen test, SOC 2 readiness.
blocked: vendors + D-V10

v0.38 → v1.0
Listing and pilots
Security review, listing, three paid pilots, GA.
blocked: D-V3 + all above

Why the stop is deliberate rather than a stall

Every one of these nine needs something that cannot be written: a ruling, a carrier contract, a Meta business account, a deployed service, or a developer org. or a ruling on whether to stay on an API that retires in 2028. v0.28.0 deliberately spent itself on the work that is valid whichever way those rulings go — hardening, encryption, isolation, audit integrity, operations — so that none of it has to be redone after the strategy is chosen.

Figure 4 — Release history and what comes next. Above the line: fifteen numbered releases between 13 and 20 September, grouped by what each phase achieved, every one verified from its own zip. Below it: nine planned releases, each carrying the decision or external dependency that blocks it. The stop is deliberate — v0.28.0 spent itself on work that stays valid whichever strategy is chosen.

7 · The decisions that unblock everything else

Use this figure as the agenda for the S13 decision gate. It is the shortest path from "we are blocked" to "here is the list, and here is what each item releases".

The eight decisions that block the next release — and exactly what each unlocks

Nothing in the right-hand column can start until the matching decision on the left is made. All eight sit with Senthil; the packet for each is already written.

D-V9 **Will AppExchange accept a net-new Open CTI listing at all?** Raise a partner support case. **SENTHIL**	→	Decides whether option (C) — finishing on Open CTI for a cash window — exists as a choice. Blocks D-V8.
D-V8 **Strategy: (A) re-platform to Partner Telephony, (B) pivot to an AI layer, (C) finish on Open CTI** Pick one. Packet written. **SENTHIL**	→	Selects the whole shape of S15-S29. Under (B) the carrier, the package and half the roadmap disappear.
D-V5 **Which telephony carrier?** Decide on landed cost per minute. **SENTHIL**	→	Unblocks the adapter, the inbound webhook, recording, call quality — and sets the pricing floor in billing.ts.
D-V6 **WhatsApp: build it, or drop the ambition?** The claim is already withdrawn. **SENTHIL**	→	Unblocks the one capability the review found genuinely differentiated and unbuilt (S21-S22, 72 points).
D-V7 **What are the real cost and funding assumptions?** ARPU contradicts the ICP by 2.5-5x. **SENTHIL**	→	Unblocks Financial_Model_v2, and with it the portfolio sequencing decision D15.
D-V4 **Do the 20 dial/log stories stay counted as Done?** Both reviewers say reclassify. **SENTHIL**	→	Moves the reported position from 96/100 to 76/100. Costs nothing but a number; buys an honest board.
D-V3 **Billing route: AppExchange Checkout, or ISVforce licence + COA?** 15% either way; 25% for OEM. **SENTHIL**	→	Unblocks the ISV agreement, the LMA setup and the pricing page copy (LC-1, LC-9).
D13 **Unpark discovery: 15 conversations with Salesforce-using sales leaders** Parked portfolio-wide since 10 Sep. **SENTHIL**	→	The review calls this the root cause. It gates the S16 kill point: three paid pilots at the ratified price, or stop.

The cheapest item on this page is the one that has been parked longest

Fifteen thirty-minute conversations cost nothing but calendar time, and they decide whether any of the other seven are worth making. Every other row commits money, a contract, or months of lead time; this one commits an afternoon. Three further decisions are open but block nothing right now: D-V1 (chat vendor — the built-in engine works), D-V10 (SSO — recommendation is 'not needed, document the smaller surface') and D15 (portfolio sequencing, which follows D-V7).

Figure 5 — The decisions that unblock everything else. Eight decisions, each with an arrow to exactly what it releases. All eight sit with Senthil. The packet for each is already written — the strategy options, the carrier trade-offs, the model rebuild scope — so each row is a ruling rather than a research task.

8 · What is completed

How to use this register. Every row names evidence you can reproduce without any system we do not have. If a row cannot be reproduced from the zip alone, it does not belong in this table — it belongs in section 9.

58 / 613
test suites / tests, all passing

100%
statements, branches, functions, lines — zero pragmas

0
production dependency advisories, any severity

55 / 10
source modules / ordered database migrations

COMPLETED	WHAT IT DOES	EVIDENCE YOU CAN RUN TODAY
The in-org softphone surface	The panel, its adapter bundle and its configuration are served by the application itself. Until v0.28.0 the page the CallCenter definition pointed at was never served at all — a live utility bar would have loaded a 404 with no error anywhere.	`test/widget.test.ts` boots the real app, fetches the real URL, asserts both script tags and the injected config, and parses the CallCenter metadata to prove the path resolves to a registered route.
Identity and session lifecycle	OAuth on an External Client App with PKCE, a signed session carrying a token id and a per-user epoch, logout, revoke-all, and role assignment that kills the target's existing sessions.	`test/sessionLifecycle.test.ts`, `test/authBoundary.test.ts` — including that a pre-v0.28.0 token shape is refused rather than grandfathered.
Web hardening	Content-Security-Policy with a Salesforce-only `frame-ancestors` allowlist, HSTS, nosniff, referrer and permissions policies, a 100 kB body cap, an inbound timeout, request ids, and rate limits that hold across instances because the counter lives in the database.	`test/hardening.test.ts` pins every header on JSON, HTML, 404 and 401 responses, and proves the org budget holds across two separate app instances sharing one repository.
Encryption at rest	AES-256-GCM envelopes over both Salesforce tokens, with a versioned key ring, per-org additional authenticated data, and a rotation sweep. The trust page has claimed this since S10; it became true on 20 September.	`test/tokenCipher.test.ts` plus a conformance test on both repository implementations asserting the persisted bytes never contain the plaintext.
Tenant isolation at the database	Row-level security, forced on every org-scoped table, compared against a tenant context set inside each transaction. A deliberately misconfigured application reads nothing.	`db/checks/rls-smoke.sql` runs in CI as a non-superuser role and proves the cross-tenant read and write are both blocked with the application absent.
Tamper-evident audit trail	A sha256 hash chain per org, append-only at the database, verifiable by the customer through an endpoint, and exempt from the record cap because an audit log with a ten-thousand-row memory is not one.	`test/auditChain.test.ts` (tampered, deleted and re-ordered rows each detected at a named position) and `db/checks/audit-append-only.sql`.
Auto-logging to Salesforce (our half)	Phone matching through the platform's own phone index, then one Task per call upserted on our external id, so a retry updates rather than duplicating. More than one matching record is reported as ambiguous and written unattached — never guessed.	`test/crmWriteBack.test.ts` drives the real route through the real writer over a local HTTP server answering the shapes Salesforce really returns.
A durable, self-draining outbox	A failed CRM write queues and is retried by a leader-elected background worker with exponential backoff and a dead-letter threshold. Before v0.28.0 the queue only drained if a human pressed a button.	`test/outboxWorker.test.ts` black-holes Salesforce for ten simulated minutes and proves the queue

		drains unattended on recovery, with leadership handing over on lease expiry.
Pre-dial refusal	E.164 normalisation against fifteen country rule sets, an org blocklist and a do-not-call registry seam — all consulted before a provider would ever be asked, so a refused number never costs a carrier attempt.	`test/numbers.test.ts` — 25 cases including the negative ones, plus the admin-only routes.
Operations tooling	Structured JSON logs with secret redaction, RED metrics per route pattern, a Prometheus endpoint, an error sink, the published 99.5% expressed as an error budget with importable burn-rate rules, a real blue/green deployment script with automatic rollback, and a disaster-recovery drill script.	`test/observability.test.ts`, `test/reviewSprint.test.ts` (the workflow and script shapes are asserted, not assumed).
The claim-to-test gate	Every sentence the product publishes about its own behaviour maps to a claim id, and every claim id must be carried by a tagged test. An unbacked sentence fails the build.	`test/claims.test.ts` walks both directions over 34 claims; `docs/claims-registry.md` is generated from the same list.
Release integrity	The evidence pack is bound to the package hash, and the comparison is symmetric: a source file the evidence never saw, or a test suite on disk the run never executed, both fail the release.	`npm run evidence:check` in CI and before every deploy; the gate was executed from the packaged zip and its own false positive fixed before release.

9 · What is not completed, and exactly why

How to use this register. Sorted by what blocks it, not by size. Nothing in this table is blocked on engineering difficulty; every row is waiting on a ruling, a commercial relationship, or a system that has to be provisioned before anyone can watch it work.

9.1 · Blocked on a decision only Senthil can make

NOT COMPLETED	WHY IT IS NOT BUILT	UNBLOCKED BY
Placing a real telephone call	No telephony adapter exists. The seam an adapter must satisfy is built and tested, and the simulator behind it labels every call `simulated: true` on every surface — but nothing dials. This was left deliberately: the provider choice sets per-minute cost and therefore the pricing floor in `billing.ts`, so it is a commercial decision, and building an adapter for an unchosen provider would mean inventing one and then testing the invention.	**D-V5** — name the carrier on landed cost per minute
Inbound calls, screen pop, call-quality telemetry	All three need the same adapter plus an inbound webhook whose signature scheme and event vocabulary are provider-specific. The routing and screen-pop logic on our side already exists.	**D-V5**
Call recording and consent	Needs the carrier, a storage location and a per-jurisdiction announcement policy. The audit chain and retention sweep it would hang on are built.	**D-V5** + a storage decision
WhatsApp — the whole omnichannel half	Never built. The product accepts a message event if something else posts one, and logs it as a Task, but there is no provider client, no number onboarding, no opt-in ledger, no 24-hour window enforcement and no templates. In v0.28.0 the *claim* was withdrawn from the help centre rather than left published, so the product no longer says it does something it does not do.	**D-V6** — build it, or drop the ambition
Which platform the in-org surface targets	Everything in the Salesforce column is built on Open CTI, which is in maintenance mode, ends life on 28 February 2028, and is closed to newly created Agentforce Service orgs. Against a five-to-nine month listing lead time, a submission started today would list onto a surface with under a year of supported life. No further work was done on that surface beyond making it correct.	**D-V9** then **D-V8**
A usable financial model	The current model never reaches breakeven inside eighteen months, its $600 ARPU contradicts its own 50–500 seat ICP by a factor of 2.5–5, it has no per-minute cost line for a telephony product, and its top plan limit equals the top of the ICP with no tier to grow into.	**D-V7** — cost and funding assumptions
An honest story count	Twenty stories promising click-to-dial and automatic logging are still counted as Done. Both external reviewers, and this release, recommend reclassifying them to "Code-complete (unproven)", which moves the reported position from 96/100 to 76/100. The definition-of-done clause is already recorded; only the reclassification itself is outstanding.	**D-V4**

Any evidence that a customer wants this	Discovery interviews were parked portfolio-wide on 10 September (D13) and have never been run. Every demand claim in the workbook is a desk assumption. The second reviewer calls this the root cause of the entire review, and it is the cheapest item on the whole list.	**D13** — fifteen conversations

9.2 · Blocked on something that has to exist in the world

NOT COMPLETED	WHY IT IS NOT DONE	UNBLOCKED BY
Any live evidence at all	There is no Salesforce developer org, no deployed service and no PostgreSQL server. This single fact is why nine of twelve capability rows read "code complete, evidence none". It is not a sequencing accident and no amount of further building changes it.	O6, O7, O7b, O9, O10
Penetration test, SOC 2, DPA execution	The internal half is done — threat model, evidence pack, control-to-code map, and a DPA template that maps every GDPR Article 28(3) and 32 clause to the code that satisfies it. The external half needs a vendor, a consultant, a legal entity and a deployed target.	Vendors, legal entity
KMS wiring, metrics backend, alerting	The key ring, the metrics endpoint and the alert rules are all built and exported. Nothing unwraps a key, scrapes the endpoint or evaluates the rules, because there is no platform.	A hosting decision
AppExchange listing and paid pilots	Package source, listing copy, asset plan and submission dry-run all exist. The listing needs the ISV agreement, the billing route, the security review ($999 per submission) and five to nine months.	D-V8, D-V3, ISV agreement
The AI layer	Transcription, AI summaries and transcript-derived coaching all need call recordings, which need a carrier. Deliberately not started: building AI governance copy before a model is ever called would create exactly the class of unbacked claim the release just spent effort eliminating.	Recordings (D-V5)

The one thing to take away if you read nothing else

The gap between this product and a shippable one is not a code gap. It is five operator days against a developer org and a database, plus six decisions that cost nothing to make but cannot be made by an engineer. Six months of further engineering with the current inputs would not move a single cell in the evidence column.

10 · The five operator steps that change the picture

How to use this section. These are the shortest paths from amber to green in figures 1 and 3. Each is written as an executable step in `docs/salesforce-dev-org-guide.md`, and each one has a corresponding manual test case in the companion test workbook.

STEP	WHAT IT NEEDS	WHAT IT WOULD PROVE, AND WHAT TURNS GREEN
O6	A PostgreSQL 16 instance	That the repository conformance suite behaves identically against a real database — caps, cascades, unique indexes and the atomic counter under concurrency. Closes R-V11 and R-V13, and turns the storage and isolation rows green.
O7	A Salesforce developer org + the package deployed	That a Task really is created once and updated on retry, that the phone match finds the right Contact, and that a Salesforce outage leaves the note safe and drains afterwards. This is the single highest-value hour available.
O7b	The same org, utility bar open	That the softphone panel actually renders inside Salesforce and the adapter loads. This is the step that would have caught the defect that survived three reviews and eight releases.
O9	Live PostgreSQL, application role	That row-level security blocks a cross-tenant read with the application deliberately misconfigured, that audit rows refuse to be updated or deleted, and that a dump restores with matching checksums.
O10	A deployed service + any metrics backend	That the burn-rate alert fires and clears during a deliberate ten-minute Salesforce outage, and that the outbox drains without anyone watching. Turns the operations row green.

11 · Glossary

TERM	MEANING AS USED IN THESE DIAGRAMS
Open CTI	The Salesforce API for embedding a third-party softphone in the Lightning UI. In maintenance mode; end of life 28 February 2028.
Partner Telephony	The Service Cloud Voice model Salesforce directs third-party carriers to; the supported replacement for Open CTI.
CallCenter definition	The metadata record that tells a Salesforce org which URL to load as its softphone.
External Client App (ECA)	The OAuth client type that replaced Connected Apps, which can no longer be created as of Spring '26.
SOSL	Salesforce Object Search Language. `FIND {digits} IN PHONE FIELDS` uses the platform's own phone index, replacing an unindexed trailing-wildcard scan.
Upsert on an external id	Writing a record keyed on our own identifier, so the platform decides insert-or-update and a retry cannot create a duplicate.
Outbox	A durable queue of CRM writes that failed, drained in order by a background worker with backoff and a dead-letter threshold.
Row-level security (RLS)	PostgreSQL policies that filter every row by the tenant context set inside the transaction, so isolation is the database's job rather than the application's.
Dead letter	A queued write parked after too many failures: visible in health output, never retried automatically.
Burn rate	How fast an error budget is being consumed. A fast burn pages a human; a slow burn opens a ticket.
E.164 / DNC	The international phone number format; a do-not-call registry that must be consulted before dialling.
O-step	An operator step: work that cannot be done in code because it needs a live system.
D-Vn	An open decision recorded in the sprint tracker, with its owner and what it blocks.
F-Pn / VB-CR-n / F-Vn	Findings from the Perplexity review, the ChatGPT review, and the internal cold reviews respectively.

Sources for every status claim in this document: the code and its test suite as shipped in `voicebridge-cti-codebase-20260920_1045.zip` (58 suites / 613 tests, 100% coverage, verified from a clean extraction); the release evidence pack bound to package hash `f0f4eb4ab67d`; the sprint tracker `claude_v024_…_20260920_1045.xlsx`; and the Salesforce help article on Open CTI retirement (005306437), read on 20 September 2026.

Prepared by Claude for Senthil (KP) · MoPr11 VoiceBridge CTI for Salesforce · document v001 · 20 September 2026 12:32 BST · end of document.